

Smart Attendance Management App

- **Developed by**
- Muhammad Umair (Fall23-bsse-0034)



Riphah International University Sahiwal



Agenda:

- Introduction
- Problem & Statement
- Proposed Solution
- System Architecture and Roles
- Key Features
- Screen Recordings
- Future Enhancement



Introduction

Overview:

SAMA is a mobile-based solution designed to automate attendance recording in educational institutions and workplaces.

Core Technology:

Device Binding and QR Codes to replace manual registers.

Goal:

To eliminate paper-based errors and save administrative time.



Problem & Solution

Problems:

- Manual attendance wastes 10-15 mins lecture time.
- Proxy attendance is common.
- Paper registers are easily damaged.

Solution:

- Automated Device Binding & QR Codes.



Proposed Solution

Automated System:

A secure android mobile application.

Verification:

Uses unique identifiers like dynamic QR codes to ensure presence.

Efficiency:

The backend automatically detects registered devices and calculate attendance according to them.



System Architecture & Roles

Architecture:

Client-server model where the mobile front-end interacts with a Firebase database.

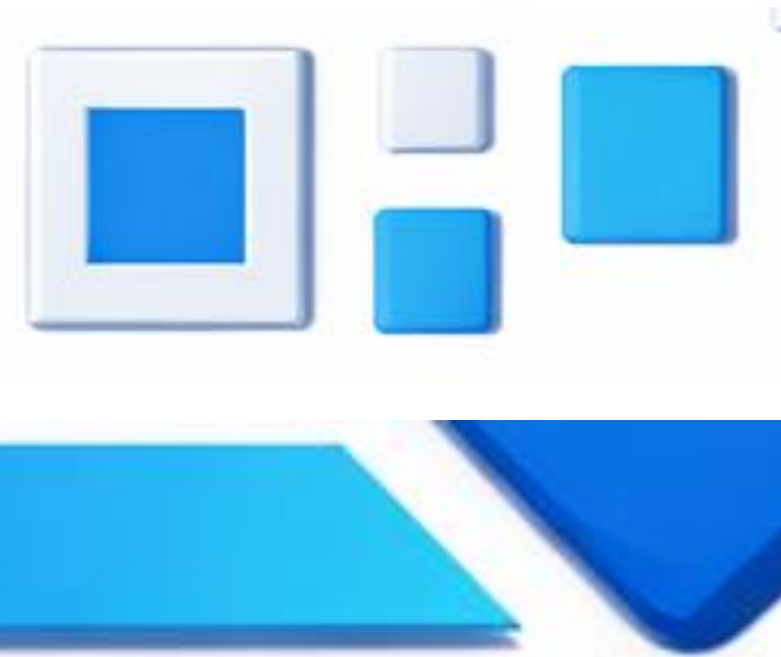
User Roles:

Admin/Teacher: Manages classes, generates sessions, and views reports.

Student: Marks presence and views personal history.

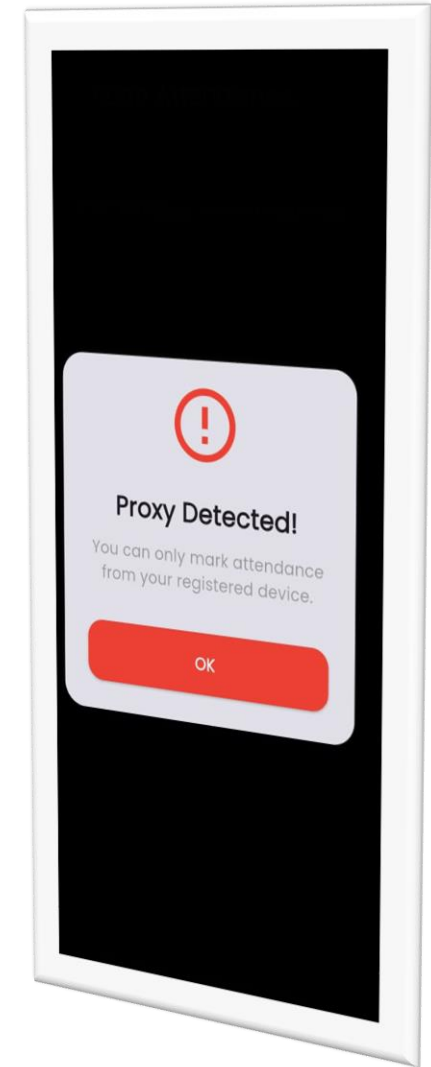
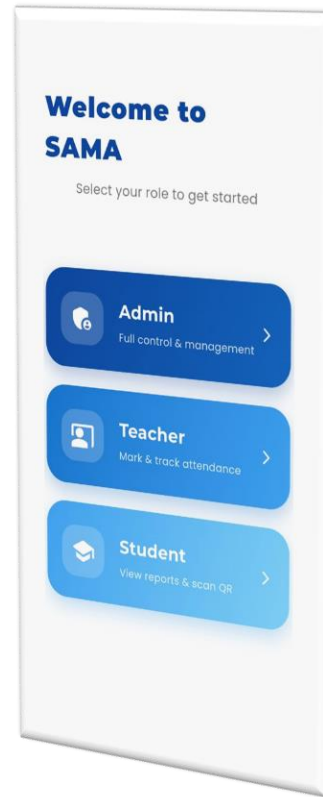
•Target Sectors:

Schools/Universities, Corporate Offices, and Events.



Key Features

- Dynamic QR Code Scanning
- Role-based Panels (Admin, Teacher, Student)
- Automated device detection



Methodology & Tech Stack

- Methodology:**

Uses the Agile Development Model for iterative testing and updates.

- Database:**

Firebase Realtime Database.

- Users Table:**

UID, Name, Role, Email.

- Attendance Table:**

User ID, Date, Timestamp, Status.

- Course Table:**

Course ID, Instructor ID, Schedule.



12:08



Login From teacher:



SAMA

Smart Attendance Management System



Developed by
TUA Software Corporation

Login From Student:



Future Enhancements

- Biometric-Integrated Authentication**

Implementing Fingerprint and Face ID verification prior to scanning the QR code. This creates a multi-layered security system, ensuring that attendance can only be marked by the actual owner of the registered device, effectively eliminating physical proxies.

- Admin Device Management & ID Reset**

Development of an administrative dashboard to manage "Device Binding." This feature allows authorized admins to securely reset a student's registered Hardware ID in cases of phone upgrades or loss, ensuring continuous access without compromising system integrity.

- Enhanced User Profiles & Manual Verification**

Upgrading the UI to include high-resolution profile pictures and detailed student analytics. This will allow teachers to perform quick visual verifications through their dashboard and provide students with a personalized experience, including attendance trend visualizations.



Any Question?

Thank You.